



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

MASTER THESIS ON SOFTWARE ENGINEERING

sửa tên

HYBRID RETRIEVAL AND LEGAL-DOMAIN ADAPTATION FOR VIETNAMESE LEGAL INFORMATION RETRIEVAL

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Acknowledgements and Declaration of Authorship

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Declaration of Authorship

I declare that this thesis is my own work and has been prepared in accordance with accepted standards of academic integrity. All sources and materials used in this research have been appropriately acknowledged and cited.

The work presented in this thesis has not been submitted, either in whole or in part, for any other degree or academic qualification at this or any other institution.

I take full responsibility for the content, accuracy, and academic integrity of this thesis.

Nghiem Phan Thanh Tuan – 2026

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Abstract



thay corpora

Vietnamese legal question answering depends on retrieving the specific provisions that support an answer. This thesis studies that retrieval problem under fixed legal corpora, recorded relevance judgments, and article-level evaluation. It asks whether lexical-semantic hybrid retrieval improves a lexical BM25 reference, whether adding multilingual-E5-base provides incremental value beyond BM25+BGE-M3, and whether legal-domain adaptation of BGE-M3 transfers from development selection to complete hybrid retrieval.

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The study evaluates ALQAC and Zalo under clean and parsed corpus representations, and introduces a Bộ Công an (BCA) question-answer collection derived from published public-service questions and answer citations. The method compares BM25, pretrained BGE-M3, pretrained E5, two-way BM25+BGE-M3 fusion, and three-way BM25+BGE-M3+E5 fusion using NDCG@10 as the primary ranking metric. BCA is evaluated with explicit coverage analysis and a component-aware protocol because many questions share legal targets.

The results show that hybrid retrieval improves the fixed BM25 reference across the reported benchmark conditions. However, the incremental value of E5 is condition-dependent: the clearest positive gain appears on parsed Zalo, while ALQAC shows small negative differences and BCA does not establish a reliable small-component advantage. The BCA analysis also shows that corpus coverage is a substantive limitation: some targets are absent or only partially represented, so ranking improvements alone cannot solve every retrieval failure.

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Legal-domain adaptation gives the thesis’s most important negative result. Study A: Hybrid-Oriented Adaptation shows substantial development-selection gains, but its corrected hybrid evaluation on data unseen during training and selection shows mean NDCG@10 decreases of 0.0520 and 0.0249 for the two-way and three-way hybrids. This evaluation is post-hoc because a preliminary analysis had already exposed the evaluation fold. Study B: Dense-Oriented Adaptation shows that fine-tuning can improve the dense encoder itself, increasing mean NDCG@10 by 0.0189, but this improvement largely disappears after integration into the full hybrids. The implication is that improving a dense retriever does not necessarily improve a complete lexical-semantic legal retrieval system.

Keywords: Vietnamese legal retrieval; BM25; BGE-M3; multilingual-E5; hybrid retrieval; contrastive fine-tuning; NDCG; legal-domain adaptation.

List of Abbreviations

Terminology follows the retrieval literature and the task definitions introduced in Chapters 3–4 [1, 2, 3, 4].

Abbreviation	Meaning	Explanation
ALQAC	Automated Legal Question Answering Competition	A benchmark and competition for evaluating legal question answering and information retrieval systems.
BCA / MPS	Bộ Công an / Ministry of Public Security collection	A legal question–document collection derived from public questions and published answers associated with Vietnam’s Ministry of Public Security.
BM25	Best Matching 25 lexical ranking function	A term-based retrieval method that ranks documents according to query-term matches, term frequency, and document length.
BGE-M3	BGE-M3 multilingual embedding model	A multilingual embedding model used in this thesis as a dense retrieval component.
CV	Cross-validation	An evaluation procedure that partitions data into subsets for model selection and performance estimation.
E5	E5 text-embedding model family	A family of dense embedding models; this thesis evaluates the <code>multilingual-e5-base</code> variant.
FT / PT	Fine-tuned / Pretrained	Distinguishes a task-adapted model from its original pretrained version.
IR	Information Retrieval	The task of retrieving and ranking documents that are relevant to a user’s information need.
MAP	Mean Average Precision	A ranking metric that averages precision across relevant results and then across queries.
MRR	Mean Reciprocal Rank	A ranking metric based on the reciprocal rank of the first relevant result for each query.
NDCG	Normalized Discounted Cumulative Gain	A ranking metric that rewards relevant documents appearing higher in the ranked result list.

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Abbreviation	Meaning	Explanation
OOF	Out-of-fold	Predictions or results produced for samples excluded from the corresponding fitting fold.
RAG	Retrieval-Augmented Generation	An architecture that retrieves external information before or during text generation.
RQ	Research Question	A specific question that the thesis investigates empirically.
SD	Standard Deviation	A statistical measure describing dispersion around the mean.

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Chapter 1

Introduction

1.1 Research Context

Retrieval-Augmented Generation (RAG) combines a language model with external evidence retrieved from a **corpus** [4]. In legal applications, the quality of this evidence is especially important because a generated answer may appear fluent while relying on an incomplete or incorrect **provision**. This thesis therefore studies the retrieval component rather than the **generation component**. Given a Vietnamese legal question, the system must rank legal articles or indexed legal documents so that the provisions relevant to the question appear near the top of the result list.

Vietnamese legal retrieval is difficult for several reasons. A user may describe a legal problem in everyday language, while the relevant article is written in formal **statutory language**. Different articles in the same law may discuss similar actors, obligations, or procedures while differing in legally decisive details. In addition, corpus construction can limit retrieval before ranking begins: if a relevant target is absent or mapped incorrectly, no scoring model can return it. These characteristics make legal retrieval a useful setting for studying both model effectiveness and evaluation design.

BM25 remains a strong **lexical reference** because legal questions often contain terms, document numbers, or phrases that also occur in relevant provisions [5, 6]. **Dense retrieval** addresses a different need: it can rank texts that are semantically related even when exact wording differs. This thesis evaluates BGE-M3 and multilingual-E5-base as pretrained dense retrievers [2, 3]. It also evaluates hybrid systems that combine lexical and dense scores. The two-way hybrid combines BM25 and BGE-M3, while the three-way hybrid adds E5.

The central problem is not whether neural retrieval is generally promising. The problem is which claims are supported when these methods are evaluated on fixed Vietnamese legal collections with recorded relevance judgments. A hybrid may beat BM25 while an added dense model contributes little beyond an already strong two-way system. A fine-tuned dense encoder may improve standalone dense retrieval while failing to improve the complete hybrid. This thesis treats such distinctions as central scientific questions rather than as secondary implementation details.

1.2 Research Gap and Approach

Prior work establishes the usefulness of lexical baselines, dense retrieval, **hybrid fusion**, and **legal-domain adaptation**, but it does not determine their behavior on the Vietnamese legal collections studied here. The empirical gap is therefore local and methodological: **the thesis needs controlled comparisons between BM25, BGE-M3, E5, two-way fusion, three-way fusion, and legal-domain adaptation under consistent evaluation boundaries.**

The study uses **three evaluation families**. First, ALQAC and Zalo are evaluated under clean and **parsed corpus** representations. These paired conditions make it possible to observe how retrieval conclusions change when the indexed text representation changes, without reducing the comparison to a single pooled score. Second, the BCA collection is constructed from published Bộ Công an questions and answers and evaluated with explicit target-coverage and component-aware analysis. Third, BGE-M3 adaptation is evaluated at both the **dense-retriever level and the complete-hybrid level**, preserving the distinction between development selection and evaluation on **unseen data**.

The experimental narrative follows a scientific sequence: define the retrieval problem, evaluate pretrained lexical and dense components, measure the incremental value of E5, **diagnose coverage and ranking failures**, test legal-domain adaptation, and interpret the results within their limits. Low-level reproduction details are retained in the appendices, while the main chapters focus on method, results, and interpretation.

1.3 Research Objectives

The main objective is to determine the effectiveness and limits of **lexical-semantic hybrid retrieval** and **legal-domain BGE-M3 adaptation** for Vietnamese legal information retrieval. This objective is divided into four operational aims.

First, the thesis compares lexical retrieval, dense retrieval, and **lexical-semantic fusion** using the same **query populations and relevance judgments**. **Second**, it measures whether **E5 adds value beyond a BM25+BGE-M3 hybrid**. Third, it evaluates whether legal-domain fine-tuning selected on development data improves retrieval on **unseen data**, both as a dense encoder and as part of the complete hybrid system. Fourth, it constructs and evaluates the BCA collection with explicit attention to corpus coverage, citation mapping, and query dependence.

The **primary ranking endpoint is NDCG@10**. **Secondary** metrics, including Hit@10, MRR@10, and MAP@10, are used to interpret **changes in first-hit** behavior and multiple-target ranking. No objective assumes a favorable result. Negative, condition-dependent, or non-transfer findings are treated as valid outcomes when they follow from the evaluation.

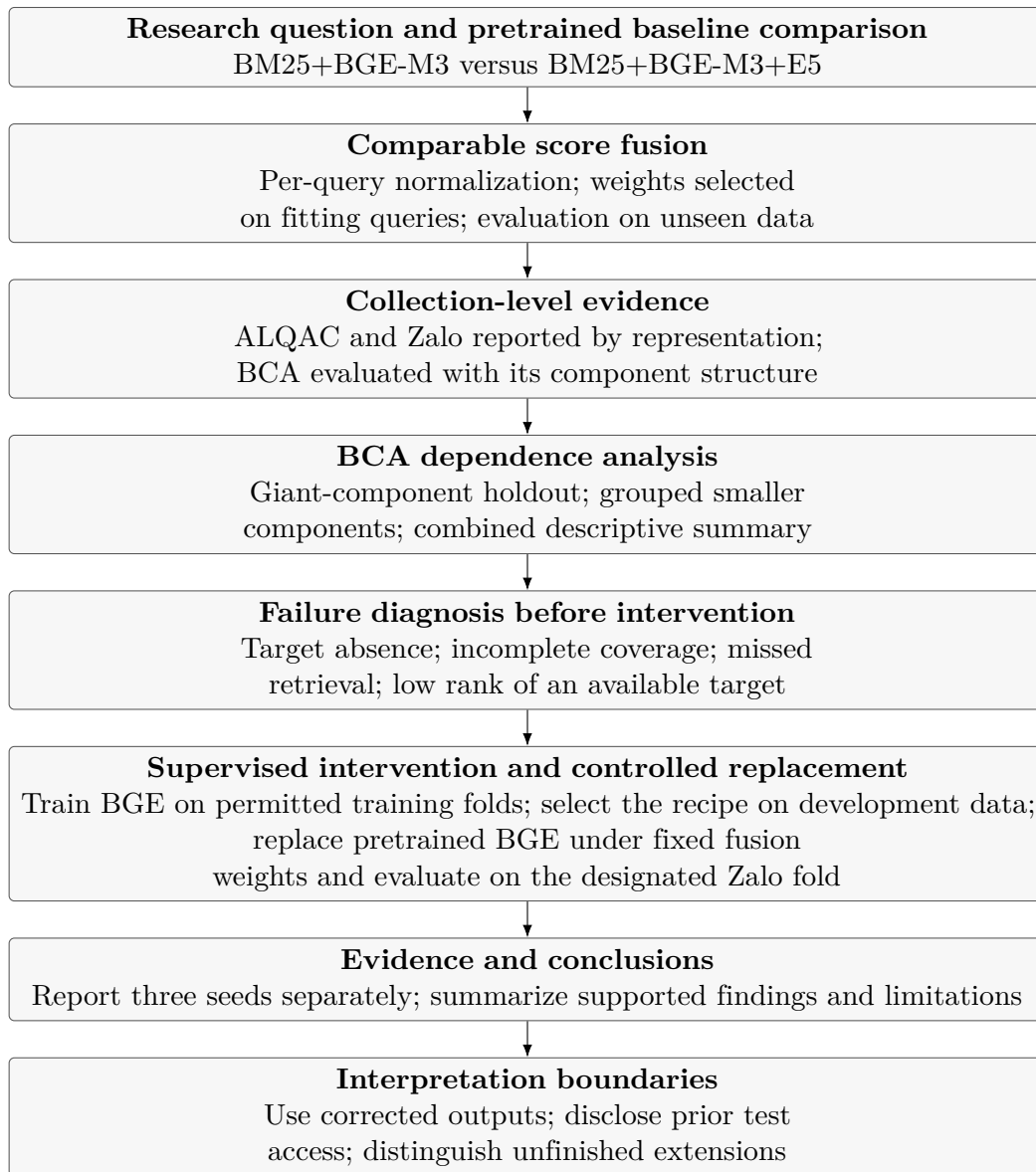


Figure 1.1: Research investigation overview. Related branches are grouped for readability; the BCA component analysis applies only to BCA. Development data select the adaptation recipe rather than replacing the protected training/evaluation split.

1.4 Research Questions

RQ1. What improvement over lexical retrieval is supported by lexical–semantic hybrid retrieval on the evaluated Vietnamese legal collections?

This question compares BM25 with pretrained dense retrievers and hybrid systems. It asks whether hybrid retrieval improves the fixed lexical reference under the evaluated collection and corpus conditions, without assuming that every hybrid must dominate every standalone retriever.

RQ2. Does adding pretrained multilingual-E5-base improve BM25+BGE-M3 across legal collections and corpus conditions?

This question isolates the incremental value of the second dense retriever. The appropriate comparator is the two-way BM25+BGE-M3 hybrid, not BM25 alone. The answer is reported by collection and corpus representation because a gain in one condition does not establish a general three-way advantage.

RQ3. Do development improvements from legal-domain BGE-M3 fine-tuning transfer to retrieval on unseen data and to the complete hybrid system?

This question separates three levels of evidence: development selection and corrected hybrid evaluation on unseen data in **Study A: Hybrid-Oriented Adaptation**, followed by dense-only and hybrid-transfer evidence in **Study B: Dense-Oriented Adaptation**. It asks whether fine-tuning improves the model in isolation and whether any dense improvement remains useful after integration with BM25 and E5.

1.5 Research Contributions

The first contribution is a **controlled empirical evaluation** of lexical–semantic hybrid retrieval for Vietnamese legal information retrieval. The thesis compares BM25, BGE-M3, E5, BM25+BGE-M3, and BM25+BGE-M3+E5 across ALQAC, Zalo, and BCA conditions. It shows that hybrid retrieval improves the fixed BM25 reference in the evaluated settings, while E5 provides condition-dependent incremental value, with the clearest gain on parsed Zalo.

The second contribution is an evaluation of legal-domain BGE-M3 **adaptation** at both dense-retriever and **complete-hybrid levels**. The thesis preserves a central negative result: Study A’s development gains do not persist in corrected hybrid evaluation on unseen data, and Study B’s dense-only improvement largely disappears after fusion. This provides evidence that improving a dense encoder does not necessarily improve a complete hybrid retrieval system.

The third contribution is a dataset and evaluation contribution centered on the BCA collection. The thesis constructs a retrieval collection from published public-service questions

and answer citations, analyzes target coverage, applies component-aware evaluation, and reports limitations caused by unavailable or partially available evidence. The contribution lies in the dataset construction, evaluation design, and reproducible comparison boundaries, not in presenting routine provenance records as algorithmic novelty.

1.6 Scope of the Thesis

The thesis evaluates retrieval effectiveness for Vietnamese legal question answering. It does not evaluate generated legal answers, legal advice quality, user satisfaction, document currency, deployment latency, or professional legal sufficiency. A higher retrieval score is interpreted as better ranking against recorded relevance judgments, not as proof that a downstream answer would be legally correct.

The evaluated collections are Vietnamese legal retrieval collections. The term “multilingual” describes the pretrained model family, not an evaluation across languages or jurisdictions. Results are therefore not generalized to other legal systems, future legal corpora, or downstream RAG systems without additional evidence.

Legal relevance is operationalized through the available annotations and extracted targets. This boundary is important for BCA: published answer citations support retrieval targets, but they are not treated as exhaustive expert relevance judgments. Queries with absent targets remain part of the analysis because they reveal a practical limitation of retrieval over an incomplete corpus.

1.7 Thesis Organization

Chapter 2 reviews related work on legal retrieval, lexical baselines, dense retrieval, hybrid fusion, fine-tuning, and evaluation design. Chapter 3 defines the theoretical background and metrics used throughout the thesis. Chapter 4 describes the retrieval pipeline, corpus construction, fusion selection, BCA construction, and adaptation methods.

Chapter 5 presents the experimental evaluation. It answers the three research questions, analyzes BCA coverage and component structure, summarizes failure modes, and develops the full adaptation narrative from development gains to regressions on unseen data and dense-to-hybrid transfer. Chapter 6 synthesizes the findings, states the contributions, discusses limitations, and identifies future work. The appendices preserve detailed search spaces, metric tables, failure cases, configurations, and reproduction information.

Chapter 2

Literature Review

The research gap concerns the value of additional retrieval capacity under legal collection and evaluation constraints. This review groups prior work by the problem it addresses: lexical evidence, learned semantic matching, fusion, adaptation, and benchmark design. Mathematical definitions are developed separately in Chapter 3. Findings on other languages, jurisdictions, and retrieval units motivate experiments here; their numerical outcomes are not transferred to Vietnamese article retrieval.

2.1 Legal Retrieval and Evidence

Legal information retrieval identifies and ranks legal materials in response to an information need. The retrieval unit may be a judgment, statute, article, paragraph, or other passage. These units define different tasks. Finding a precedent relevant to a case is different from identifying an article that supplies the rule needed to answer a short question. COLIEE makes this distinction explicit by separating case retrieval, case entailment, statute retrieval, and statute entailment. Its statute tasks use Japanese bar-examination questions; its case tasks concern Canadian case law. Goebel et al. [7] therefore provides a useful taxonomy, rather than a directly interchangeable benchmark for Vietnamese law.

For this thesis, relevance is operationalized through the dataset’s question-to-provision judgments. The ALQAC 2024 organizers define relevant articles as those that can be used to answer the question. The competition distinguishes document retrieval from question answering and describes its data as manually annotated Vietnamese statute-law questions. This task definition supports evaluating retrieval independently from answer generation. It does not imply that every retrieved article constitutes a complete answer or that the available judgments exhaust every legally defensible supporting provision [8].

The distinction between semantic association and evidential relevance is fundamental to the analysis developed here. Consider an illustrative question about a deadline for submitting an application. A document discussing the same application may receive a high semantic score while describing a different procedural stage. Conversely, a short provision giving the required deadline may share few words with a conversational question. This example is an analytical illustration, not an observed project failure. It explains why both lexical and semantic signals deserve evaluation: one may preserve distinctive terminology,

while the other may connect different formulations of the same information need. Neither signal explicitly proves that all conditions of the question are satisfied.

Article-level retrieval also introduces a boundary decision. An article may contain a rule, several exceptions, and cross-references. Dividing it into smaller passages can make a particular sentence easier to match, but the resulting passage may omit context necessary to interpret it. Retaining the complete article preserves context but forces the ranking representation to accommodate several subtopics. There is no representation-independent answer to this tradeoff. If the ground truth identifies articles, passage-level predictions must be mapped back to articles before effectiveness is compared. Otherwise, a change in retrieval unit is confounded with a change in the number and definition of relevant items. Vietnamese preprocessing requires similar care. PhoBERT explicitly uses Vietnamese word segmentation before subword processing and evaluates language-specific NLP tasks [9]. Its evidence establishes that segmentation is a substantive modeling decision in Vietnamese NLP; it does not establish that a particular segmenter improves this thesis’s retriever. The implication for lexical retrieval is that whitespace units and segmented words define different vocabularies and frequency statistics. The implication for dense retrieval is different: its input should follow the encoder’s documented preprocessing contract. Applying a lexical segmenter’s output to an encoder without checking that contract would introduce another experimental variable.

The broader motivation includes retrieval-augmented generation. Lewis et al. combine a parametric generator with retrieved nonparametric information and evaluate knowledge-intensive NLP tasks [4]. This establishes an architectural connection between retrieval and generation. The present thesis nevertheless treats retrieval effectiveness as a separate construct. A ranking metric describes the ordering of judged evidence; it does not directly measure whether a generator interprets that evidence correctly, cites it faithfully, or produces an appropriate legal answer. Any downstream benefit inferred from retrieval improvements must therefore remain a hypothesis until generation is evaluated.

2.2 Lexical Retrieval as a Serious Reference

The lexical family ranks documents using observable term evidence. Its attraction in legal retrieval is the direct relationship between score contributions and the indexed wording. Robertson and Zaragoza [5] develop the probabilistic relevance framework underlying BM25; Rosa et al. [6] provide a legal case-retrieval example in which a vanilla BM25 submission is competitive. These sources play different roles: the former explains a scoring framework, while the latter demonstrates that a lexical comparator cannot be dismissed simply because neural methods are available.

The case-retrieval finding does not settle article retrieval in Vietnamese. A named legal

concept may be expressed differently in a conversational question, and a matched term may occur in a provision addressing a different procedural condition. The relevant gap is therefore whether a learned signal adds useful ordering evidence under the same candidate collection and judgments. A weakly specified BM25 baseline would make that question difficult to answer because preprocessing and article boundaries affect its vocabulary and document statistics.

This thesis retains BM25 as the fixed lexical reference. It does not introduce word segmentation or a new lexical representation as a demonstrated contribution without a matched experiment. The mathematical treatment of term saturation and length normalization appears in Section 3.2; the actual tokenizer and implementation are specified in Section 4.4.1.

2.3 Multilingual Neural Retrieval

The neural family replaces explicit vocabulary coordinates with learned representations. Sentence-BERT and Dense Passage Retrieval provide influential approaches to separate query/document encoding and supervised retrieval [10, 11]. Their methodological relevance is that document representations can be prepared before a query arrives. Their evaluation tasks do not independently establish effectiveness on the legal collections studied here.

BGE-M3 and multilingual E5 extend the motivation to multilingual retrieval [2, 3]. The BGE-M3 paper describes multiple retrieval functions, while the E5 technical report describes a model family with distinct checkpoints and training configurations. The present study uses BGE-M3 dense vectors and E5-base as separately identified channels. This restriction matters: a result for one dense configuration cannot support a conclusion about all model modes or all members of a model family.

A single-vector representation supplies an efficient separation between document encoding and query-time comparison, but it compresses the document before seeing the query. ColBERT explores a different design through contextualized token representations and late interaction [12]. That alternative identifies a limitation of the studied design space; it is not an evaluated solution to this project’s observed failures.

These studies justify testing multilingual representations while leaving two local questions unresolved: whether their rankings improve on the lexical reference, and whether their disagreements are useful after fusion. The latter cannot be answered by a standalone leaderboard position. It requires comparing the complete hybrid against the hybrid it extends.

2.4 Lexical–Semantic and Multi-Retriever Fusion

Fusion studies address how heterogeneous rankings or scores can be combined. Cormack et al. [13] introduce reciprocal rank fusion, while Bruch et al. [14] analyze lexical–semantic fusion including convex score combination. The distinction is substantive: rank fusion uses positions, whereas weighted score fusion retains information about score distances after a specified normalization. Their behavior depends on candidate policy and parameter selection; the literature does not justify a universal ordering independent of those settings. The hypothesis behind lexical–semantic fusion is complementarity. Lexical evidence and learned similarity may recover different relevant items or order the same items differently. Adding a second dense retriever poses a stricter question: it must contribute beyond a system that already combines lexical and semantic information. Strong standalone performance is insufficient if the added channel repeats evidence already supplied by the other components [14].

The project tests that stricter question by comparing BM25+BGE-M3 with a search space that additionally permits E5-base. Because the larger weight simplex includes zero-weight boundaries, selection can ignore the new channel. An improved selection score could reflect a wider search rather than stable additional evidence. The evaluation must therefore inspect rankings on unseen data and condition-specific changes, not only selected weights. This reasoning motivates RQ1 and RQ2 while defining their limits. The thesis evaluates the implemented score fusion; Reciprocal Rank Fusion, alternative normalization, and candidate-depth changes remain related approaches unless matched experimental outputs support a comparison.

2.5 Legal-Domain Adaptation and Hard Negatives

Domain adaptation can concern language modeling or retrieval supervision. LEGAL-BERT studies legal language representations, whereas Dense Passage Retrieval trains query–passage matching [15, 11]. Legal text exposure alone therefore does not identify the retrieval objective optimized by a model. For this thesis, the intervention is supervised contrastive adaptation of a dense encoder using query–article groups.

Hard-negative research examines how informative competitors affect retrieval training. Zhan et al. [16] investigate training strategies that change the source or refresh of difficult examples. This makes the mining procedure a substantive part of the experiment: the source retriever, corpus, and filtering rules determine the distinctions the learner is asked to acquire. In legal collections, excluding known positive identifiers is necessary but does not establish that every remaining competitor is legally irrelevant.

The system-level question remains after training. An adapted dense model may change its

overlap with BM25, so its value inside a hybrid cannot be inferred from training loss or from a separate dense-only score. Fixed fusion weights test replacement of one component; retuning weights on permissible selection data tests an additional system adjustment. Neither design should be confused with selecting weights from final test outcomes.

RQ3 addresses the completed fixed-weight hybrid experiments. The hypothesis is that development-selected adaptation supplies useful ranking changes outside the selection cohort. Study B: Dense-Oriented Adaptation measures both dense-only and locked-weight hybrid outcomes, allowing the thesis to distinguish encoder improvement from system improvement. Study B’s different recipe and previously observed test remain explicit in Section 5.6.3.

2.6 Benchmarks and Evaluation Design

The literature supplies several kinds of evidence, each answering a different question. ALQAC specifies a directly relevant Vietnamese statute-retrieval task. COLIEE distinguishes retrieval from entailment and supplies legal benchmarks in other jurisdictions. BEIR studies generalization across heterogeneous retrieval collections. Model papers establish representation methods, while training studies investigate optimization. A defensible related-work synthesis separates these evidential roles rather than treating every reported score as a comparable estimate of legal retrieval quality [8, 7, 17].

Table 2.1: Prior research grouped by task, intervention, and evaluation role.

Source	Task and language scope	Representation or intervention	Evaluation emphasis	Relevance and boundary for this thesis
ALQAC Organizers [8]	Vietnamese statute articles retrieved for questions	Competition task definition	Identifying answer-supporting articles	Direct task motivation; project-specific corpus versions still require local provenance
Goebel et al. [7]	Canadian case law and Japanese statute tasks	Multiple competition submissions	Task-specific retrieval and entailment evaluation	Defines legal task distinctions; jurisdiction and retrieval unit differ
[6]	COLIEE legal case retrieval	Vanilla BM25 submission	Competition retrieval outcome	Supports a serious lexical baseline; does not establish Vietnamese article performance

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Source	Task and language scope	Representation or intervention	Evaluation emphasis	Relevance and boundary for this thesis
Thakur et al. [17]	Heterogeneous general retrieval collections	Lexical, dense, sparse, reranking, and late interaction	Zero-shot transfer across tasks	Motivates reporting generalization separately from development gains
Chen et al. [2]	Multilingual and multiple-granularity retrieval	M3-Embedding	Model benchmark evaluation	Establishes BGE-M3 capabilities; dense-only deployment is a restricted configuration
Wang et al. [3]	Multilingual embedding and retrieval tasks	E5 model family	Training recipe and benchmark evaluation	Supplies E5 context; base, large, and instruction-tuned variants remain separate
[14]	Lexical-semantic fusion experiments	Convex combination and Reciprocal Rank Fusion	Fusion behavior and tuning	Motivates explicit fusion specification; local gains need local evaluation
[16]	Dense retrieval training	Hard-negative strategies	Ranking optimization	Motivates careful negative construction; legal label validity remains a separate issue

The table synthesizes the cited primary sources; it deliberately omits incompatible numerical leaderboards. For example, a case-retrieval F-measure and article-level NDCG at a fixed cutoff summarize different outputs. Comparing their magnitudes would create an apparent ranking of methods without a common task, relevance set, or metric. The appropriate transfer from prior work is a methodological argument: include a strong lexical comparator, distinguish representation from scoring, and evaluate model selection independently from final performance.

BEIR is particularly useful for that argument. Thakur et al. introduce heterogeneous zero-shot retrieval evaluation and report that BM25 remains robust, while reranking and late-interaction approaches offer strong average effectiveness with additional computational cost. They also identify generalization limitations of evaluated dense and sparse systems [17]. This does not mean that dense retrieval cannot generalize. It means that in-domain improvement is insufficient evidence of transfer. A thesis about multilingual legal adaptation should report each dataset and its selection status rather than allow an aggregate to conceal different outcomes.

Ranking metrics introduce another layer of task definition. Discounted cumulative gain was developed to account for relevance gains at different ranking positions [18]. In the usual normalized construction, observed discounted gain is divided by the gain of an ideal

ranking under the same judgments. The interpretation depends on the labels. Binary article judgments evaluate the order of known relevant items; they do not produce a scale of legal importance simply because the metric can also accommodate graded labels.

The difference between finding any relevant article and retrieving the complete labeled evidence set is equally important. For a query with several relevant articles, a hit indicator becomes one as soon as one is found. Recall instead depends on how many labeled relevant articles are recovered relative to their total number. The two coincide for a single-positive query but need not coincide otherwise. This is a consequence of their definitions and explains why legacy metric names must be checked against evaluator code before being used in a comparison with the literature [1].

Legal benchmarks also require attention to unavailable targets. If a labeled article is absent from the indexed corpus, a ranking model cannot retrieve that identifier. Evaluating only answerable questions estimates ranking quality conditional on corpus coverage. Evaluating every question estimates the combined outcome of coverage and ranking. These are different estimands, and neither should silently replace the other. This distinction is especially relevant when an experimental representation is obtained by extraction or mapping, because the data pipeline can change whether a target is represented at all.

Related architectures provide boundaries rather than promised solutions. A cross-encoder reranker can evaluate a query and candidate jointly, while ColBERT retains token-level matching [10, 12]. Such methods are relevant alternatives when a single-vector representation loses a distinguishing condition. Their presence in the literature does not establish that they would resolve the specific failures observed here. A future comparison would need fixed candidate sets, comparable relevance judgments, and explicit computation measurements.

2.7 Research Gap and Positioning

This thesis is positioned as a controlled empirical investigation of Vietnamese legal retrieval. Its purpose is not to introduce BM25, multilingual embeddings, contrastive learning, or linear fusion as new algorithms. The contribution lies in determining what these components accomplish under the studied legal collections, representation conditions, and evaluation constraints. The literature makes the comparisons worthwhile; the experiments determine which conclusions can be sustained.

Three distinctions organize that investigation. First, lexical-semantic combination is separated from the addition of another dense channel. Demonstrating that a hybrid improves on BM25 would not establish that every further retriever adds useful evidence. Second, dense adaptation is separated from hybrid adaptation. An improvement in the dense component would not by itself establish an improvement in the combined ranking.

Third, development performance is separated from performance on unseen data and across datasets. A configuration selected using a particular collection cannot be described as unselected generalization evidence on that same collection.

Representation conditions are another substantive part of the study. The clean or parsed text supplied to a model is part of the experimental treatment, as are article boundaries and target mappings. Treating these merely as preprocessing details would obscure why a model gains or loses evidence. At the same time, the thesis should not attribute every score change to parser quality when candidate coverage, text content, and normalization populations may change together. The empirical analysis therefore distinguishes directly measured outcomes from explanations that require further intervention or manual inspection.

The reviewed sources support hypotheses about lexical-semantic complementarity, multi-lingual representation, and the value of difficult training examples. They do not warrant assuming that all three hypotheses will succeed simultaneously. A limited contribution from E5, a dense-only improvement without hybrid benefit, or a development gain that fails to transfer can each answer a valid research question. Negative findings are particularly informative when the comparison fixes the corpus, evaluation, and model identities sufficiently to make the failure interpretable [14, 2, 16].

Chapter 4 translates the theoretical definitions into a concrete retrieval pipeline. It specifies the indexed units, query processing, component scores, normalization population, and fusion selection. The experimental evaluation can then interpret an observed difference as evidence about a defined system rather than about an ambiguous model name. Where a fine-tuning evaluation has not been completed, the thesis preserves that boundary instead of turning theoretical motivation or training progress into an empirical result.

2.8 Chapter Summary

Prior work establishes relevant model families and evaluation principles, but it does not determine their incremental value on these particular Vietnamese legal collections. The thesis therefore asks three increasingly restrictive questions: whether fusion improves a lexical reference, whether another dense channel improves an existing hybrid, and whether development-selected adaptation improves the hybrid on unseen data. Chapter 3 defines the concepts needed to formalize these comparisons.

Chapter 3

Theoretical Background

This chapter defines the retrieval models, fusion methods, training objectives, and evaluation metrics used in the thesis. Chapter 4 specifies their implementation.

3.1 Queries, Articles, and Relevance

Let D be an ordered article corpus, Q a query population, and R_q the deduplicated relevance targets for query q . Retrieval returns ranked article identities, not generated answers. Corpus coverage determines whether targets are available; ranking determines their positions. No scorer can recover a target absent from the corpus [1].

Targets may identify an article or an entire law. Because one law-level annotation need not represent multiple independent article judgments, evaluation credits each target at most once.

3.2 Lexical Scoring and BM25

3.2.1 TF-IDF as a Conceptual Starting Point

Sparse representations weight indexed terms. TF-IDF rewards within-document occurrences while reducing the influence of terms common across the collection. It provides the conceptual basis for BM25, not an additional experimental baseline in this thesis [5].

3.2.2 BM25

A common single-field BM25 score is [5]

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1 (1 - b + b|d|/\text{avgdl})}.$$

Here, $f(t, d)$ is term frequency, $|d|$ is document length in indexed tokens, avgdl is average collection length, and $\text{IDF}(t)$ weights term rarity. Parameters k_1 and b control frequency saturation and length normalization. IDF conventions vary by implementation; Chapter 4 identifies the version used [5].

Term contributions saturate, limiting the benefit of repetition. Length normalization also makes scores sensitive to parsing: removing unrelated text can increase a score, whereas removing a matching term eliminates its contribution. Parsed text therefore need not lower every lexical score.

BM25 is a substantive baseline: a vanilla submission placed second in COLIEE 2021 case retrieval [6]. However, lexical matching cannot directly recover unrepresented paraphrases, and matching terms alone do not establish legal applicability. **Dense retrieval tests whether learned representations supply complementary evidence** [5].

3.3 Dense Representations and Similarity

Bi-encoders map queries and documents separately to continuous vectors, allowing document embeddings to be precomputed. **Sentence-BERT and Dense Passage Retrieval** provide influential examples of learned text matching, although their original tasks differ from Vietnamese legal retrieval [10, 11].

For query and document encoders E_q and E_d , let $u = E_q(q)$ and $v = E_d(d)$ belong to $\mathbb{R}^m$. Common similarity scores are

$$s_{\text{dot}}(q, d) = u^\top v, \quad s_{\text{cos}}(q, d) = \frac{u^\top v}{\|u\|_2 \|v\|_2}.$$

Dot product equals cosine similarity for unit-normalized nonzero vectors. Encoder revision, pooling, normalization, and truncation nevertheless remain distinct parts of the representation [10, 11].

A single document vector summarizes information for many possible queries. Token-level late-interaction models such as ColBERT offer a different design; results for a single-vector encoder do not characterize all neural retrieval architectures [12].

3.3.1 BGE-M3

BGE-M3 supports multilingual dense, sparse, and multi-vector retrieval, with inputs up to 8,192 tokens [2]. This thesis uses its dense representation. Combining BM25 with that representation is distinct from combining BGE-M3’s own retrieval modes, and the model’s supported input length should not be confused with the input length selected for a particular experiment.

BGE-M3 provides a multilingual semantic channel whose effectiveness and complementarity with BM25 must be measured on the target collections rather than inferred from model capabilities.

3.3.2 Multilingual-E5

The multilingual E5 technical report describes contrastive pretraining followed by supervised fine-tuning [3]. The checkpoint used here, `intfloat/multilingual-e5-base`, has 12 layers, 768-dimensional embeddings, attention-mask-aware mean pooling, normalized vectors, and a 512-token input limit. Its retrieval convention prefixes queries with `query:` and documents with `passage:` , including non-English text [3].

E5-base supplies a second dense signal. Its incremental value must be assessed against BM25+BGE-M3, since different encoders may make similar errors. Differences in tokenization and truncation can also change the evidence each encoder receives; score differences alone do not isolate semantic understanding.

3.4 Normalization and Fusion

3.4.1 Lexical–Semantic Hybrid Retrieval

Hybrid retrieval combines lexical and semantic evidence. Its central hypothesis is that the combined ranking improves on its components, but effectiveness depends on the fusion configuration [14].

Full-corpus fusion scores every document with every channel. Candidate-union fusion instead ranks only documents returned by first-stage retrievers; weights cannot recover a target excluded from every candidate list. Candidate depth and score combination are therefore separate design choices [14].

3.4.2 Multi-Retriever Fusion

For normalized BM25, BGE-M3, and E5-base scores $\tilde{S}_b$, $\tilde{S}_g$, and $\tilde{S}_e$, three-way fusion is

$$S(q, d) = w_b \tilde{S}_b(q, d) + w_g \tilde{S}_g(q, d) + w_e \tilde{S}_e(q, d), \quad w_b, w_g, w_e \geq 0, \quad w_b + w_g + w_e = 1.$$

Setting $w_e = 0$ recovers BM25+BGE-M3 when all other settings are unchanged. A search containing this boundary cannot have a lower optimal development score than its **two-way subset**, but improvement on **unseen data** is not guaranteed. A selected zero E5 weight indicates that the additional channel is unused.

Complementarity may affect early precision, first-hit position, or target recovery differently. Multiple ranking metrics are therefore needed; positive fusion weights alone do not establish useful complementary errors.

3.4.3 Score Normalization and Fusion

For raw channel score S_j and normalization candidate set C_q , min-max normalization is [14]

$$\tilde{S}_j(q, d) = \frac{S_j(q, d) - a_j(q)}{b_j(q) - a_j(q)}, \quad a_j(q) = \min_{x \in C_q} S_j(q, x), \quad b_j(q) = \max_{x \in C_q} S_j(q, x).$$

The expression requires $b_j(q) > a_j(q)$. Constant-score channels require a deterministic rule, such as mapping all values to zero, because they provide no within-query ordering evidence.

An alternative is z-score normalization,

$$Z_j(q, d) = \frac{S_j(q, d) - \mu_j(q)}{\sigma_j(q)},$$

where $\mu_j(q)$ and $\sigma_j(q) > 0$ are the mean and standard deviation over C_q . Unlike min-max scores, z-scores are not bounded by zero and one. Both methods depend on the candidate population: adding documents can change normalized values without changing existing raw scores.

Reciprocal rank fusion (RRF) combines positions instead of score magnitudes [13]:

$$\text{RRF}(d) = \sum_{j \in J_d} \frac{1}{c + r_j(d)}.$$

Here, $r_j(d)$ is a one-based rank, J_d contains the rankings that include d , and $c > 0$ is a smoothing constant. RRF avoids direct score-scale comparison but remains sensitive to rank depth and parameter choices. Evidence favoring tuned convex fusion in particular settings does not establish a universal ordering; normalization, weights, candidate depth, and selection data must all be specified [14].

3.5 Contrastive Adaptation and Negative Sampling

Legal-domain language-model pretraining and supervised retrieval fine-tuning are different interventions. LEGAL-BERT adapts language representations to legal text, whereas DPR learns query-passage ranking from relevance supervision. Familiarity with legal prose does not itself establish retrieval effectiveness [15, 11].

3.5.1 Contrastive Learning

A single-positive InfoNCE-style loss encourages a relevant document to outscore competing documents [19, 11]:

$$\mathcal{L}(q) = -\log \frac{\exp(s(q, d^+)/\tau)}{\exp(s(q, d^+)/\tau) + \sum_{d^- \in N(q)} \exp(s(q, d^-)/\tau)}.$$

Here, d^+ is a labeled positive, $N(q)$ is the negative set, s is similarity, and $\tau > 0$ is temperature. Temperature changes the emphasis on score differences, while batch composition determines the competitors. Lower training loss does not guarantee better retrieval on unseen data.

For multiple positives, a set-positive loss can sum positive exponentials in the numerator. Alternatively, positive-specific losses can be averaged while masking other known positives from negative positions. These objectives are not interchangeable; Chapter 4 states the loss and masking policy of each study.

3.5.2 Negative Sampling

Random negatives are sampled without a high-rank requirement; in-batch negatives reuse passages from other examples; hard negatives are high-ranking candidates without positive labels [11, 16].

Excluding known positives prevents label contradictions but does not prove that unjudged articles are irrelevant. In particular, selecting one positive must not cause another known positive to become an unmasked negative. Explicit-negative filtering, in-batch masking, and semantic review provide different safeguards and should be reported separately.

3.5.3 Hard-Negative Mining

Static mining fixes competitors before training; dynamic mining refreshes them as the model changes [16]. Reproduction requires the miner checkpoint, corpus snapshot, ranking range, filtering rules, and any refresh schedule.

Adaptation must be evaluated both alone and within the hybrid: a dense improvement may overlap with BM25’s strengths or reduce useful disagreement. Retaining pretrained fusion weights tests component substitution, whereas retuning on permissible development data tests adaptation of the complete system.

3.6 Target Matching and Evaluation Metrics

The evaluator selects ten candidates by partial partitioning and sorts them by decreasing score. Each retrieved article is checked against unmatched targets. An empty article identifier or `toanvan` denotes a law-level wildcard; otherwise, both law and article identifiers must match. Each target earns credit at most once, so retrieving several articles from one law does not multiply credit for a single whole-law target.

Let $y_{q,r}$ indicate whether rank r matches a previously unmatched target. The primary metric is [18]

$$\text{NDCG@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r} / \log_2(r+1)}{\sum_{r=1}^{\min(|R_q|, 10)} 1 / \log_2(r+1)}.$$

Queries without targets receive zero. Unavailable targets remain in the ideal-gain denominator, making coverage part of end-to-end effectiveness. Mixed law-level and article-level targets remain subject to the implemented matching order.

The hit metric is

$$\text{Hit@}k(q) = \mathbb{1} \left[\sum_{r=1}^k y_{q,r} > 0 \right],$$

reported at $k = 1, 5, 10$. It indicates whether any target is recovered, not the proportion recovered. True target recall is $\text{Recall@}k(q) = \sum_{r=1}^k y_{q,r} / |R_q|$ for nonempty R_q . Hit and recall coincide for single-target queries but can differ for multi-target queries; the canonical baseline aggregate does not report true target recall, including Recall@100 [1].

With r_q^* denoting the first matching rank,

$$\text{MRR@10} = \frac{1}{|Q|} \sum_{q \in Q} \frac{\mathbb{1}[r_q^* \leq 10]}{r_q^*},$$

where a missing match contributes zero. The implemented average-precision metrics are

$$\begin{aligned} P_q(r) &= \frac{1}{r} \sum_{j=1}^r y_{q,j}, \\ \text{AP@10}(q) &= \frac{\sum_{r=1}^{10} y_{q,r} P_q(r)}{\max(1, \min(|R_q|, 10))}, \\ \text{MAP@10} &= \frac{1}{|Q|} \sum_{q \in Q} \text{AP@10}(q). \end{aligned}$$

This truncated AP denominator may differ from other implementations. All aggregate metrics are query-macro means, including zero-score queries, unless an answerable-only cohort is explicitly stated. They measure annotated target recovery, not exhaustive legal relevance [1].

3.7 Grouped Evaluation and Statistical Interpretation

Queries can share target laws, and multi-law queries connect those laws into components. Keeping each component within one fold prevents reference-law overlap across fusion selection and evaluation. It does not partition the searchable corpus or establish that encoders have never seen the legal text.

Query-, component-, and fold-macro averages differ when group sizes differ. Confidence intervals must identify their resampling unit; variation across training seeds is not uncertainty over a new query population. Likewise, an interval spanning zero neither proves equivalence nor erases an observed difference. Chapter 5 interprets the stored statistical tests within these boundaries [20].

3.8 Chapter Summary

BM25 and bi-encoders supply lexical and semantic scores; fusion combines them, and contrastive adaptation changes the dense representation. Target-aware metrics and grouped evaluation assess these interventions under explicit corpus and selection boundaries.

Chapter 4

Proposed Methodology

This chapter defines the retrieval task, datasets, scoring methods, fusion procedure, evaluation design, and legal-domain adaptation studies. The method uses established retrieval models and a controlled evaluation pipeline; the contribution lies in the dataset construction, experimental comparison, and interpretation of results rather than in a new ranking algorithm.

4.1 Problem Formulation

Let $D = \{d_1, \dots, d_n\}$ denote an ordered corpus of legal retrieval units. Each unit contains article text and a normalized legal identifier. For a query q , the relevance set R_q contains the law–article targets recorded for that question. A retrieval configuration θ produces a ranked list

$$\pi_q(\theta) = (d_{(1)}, d_{(2)}, \dots, d_{(k)}),$$

where $k = 10$ for the primary evaluation window.

The selection objective is mean NDCG@10 on a designated selection population:

$$\theta^* = \arg \max_{\theta \in \Theta} \frac{1}{|Q_{\text{sel}}|} \sum_{q \in Q_{\text{sel}}} \text{NDCG@10}(\pi_q(\theta), R_q).$$

The candidate set Θ contains tested fusion weights and, in the adaptation studies, selected encoder-training recipes. Labels are used only for selection and evaluation. At retrieval time, the system receives the query text and the indexed corpus; it does not receive the answer, target labels, or relevance judgments.

This formulation makes two limitations explicit. First, a target absent from the searchable corpus cannot be retrieved by any ranking model. Second, retrieval effectiveness is not the same as generated legal-answer correctness. The study therefore evaluates article ranking, not downstream legal advice.

4.2 Retrieval and Evaluation Pipeline

The baseline architecture compares individual retrievers and weighted hybrids. BM25 provides the lexical score. BGE-M3 and multilingual-E5-base provide dense vector scores. The two-way hybrid combines BM25 and BGE-M3; the three-way hybrid combines BM25, BGE-M3, and E5.

Each component scores the same ordered corpus. Scores are normalized per query and combined by weighted addition. The resulting top-ten rankings are evaluated against the recorded targets. The system does not use a cross-encoder reranker, query router, query expansion module, approximate nearest-neighbor filter, or answer-generation stage. This restriction keeps the experiment focused on first-stage lexical-semantic retrieval.

Figure 4.1 summarizes the baseline scoring and evaluation architecture.

4.3 Datasets and Corpus Construction

The evaluation covers ALQAC, Zalo, and BCA. ALQAC and Zalo are each evaluated in two corpus representations: a **clean reference representation and a parsed-text representation**. The paired A/B conditions share query populations and therefore should not be treated as independent datasets. BCA is evaluated separately because its target graph contains strong dependency between questions that cite the same laws.

Table 4.1: Canonical baseline evaluation conditions.

Condition	Queries	Articles	Evaluation boundary
ALQAC A, clean	811	3,363	Law-component grouped out-of-fold evaluation over the clean reference representation
ALQAC B, parsed	811	3,381	Same ordered queries with separately constructed parsed articles
Zalo A, clean	3,196	61,425	Law-component grouped out-of-fold evaluation over the clean reference representation
Zalo B, parsed with fallback	3,196	61,425	Same ordered queries with parsed text where available and reference text retained otherwise

Continued on next page

Condition	Queries	Articles	Evaluation boundary
BCA, canonical evaluation	546	14,789	Giant-component holdout plus small-component out-of-fold evaluation

In ALQAC A, reference legal articles are flattened while preserving article identity. ALQAC B uses parsed source material; parsing changes article boundaries and can change the number of retrieval units. In Zalo A, reference articles define the clean corpus. In Zalo B, parsed text replaces reference text where an aligned parsed version is available, while fallback preserves article availability when parsing fails. Consequently, the Zalo A/B comparison evaluates a specific parsed-with-fallback construction, not the isolated effect of parsing alone.

The BCA collection is constructed from public questions and published answers associated with the Bộ Công an portal. The questions supply retrieval inputs. The published answers provide citation evidence from which legal targets are derived. The resulting collection contains 652 source records, of which 546 are retained for retrieval evaluation because they have at least one extracted law reference. **Later coverage analysis distinguishes answerable, partially answerable, and unavailable target cases.**

4.3.1 BCA Target Construction

The BCA construction process converts public question-answer records into a retrieval dataset. Question text is cleaned and retained as the query. Answer text is used only to extract law and article references; it is not appended to the query at retrieval time. Extracted law names are normalized and aligned to legal identifiers in the searchable corpus. Article identifiers are normalized separately so that matching remains identity-based rather than text-similarity-based.

Figure 4.2 summarizes the construction process. The process is useful because it produces naturally occurring legal information needs outside the competition collections. Its limitations are equally important: published answers are not exhaustive relevance judgments, citation extraction is heuristic, and some cited material is absent from the indexed corpus.

4.3.2 Worked BCA Example

Table 4.2 illustrates how one public question becomes a retrieval instance. The question asks whether apartment buildings are subject to mandatory fire-prevention equipment requirements. The published answer cites a legal instrument and relevant articles. The dataset stores the question as the query and the cited provisions as target identities.

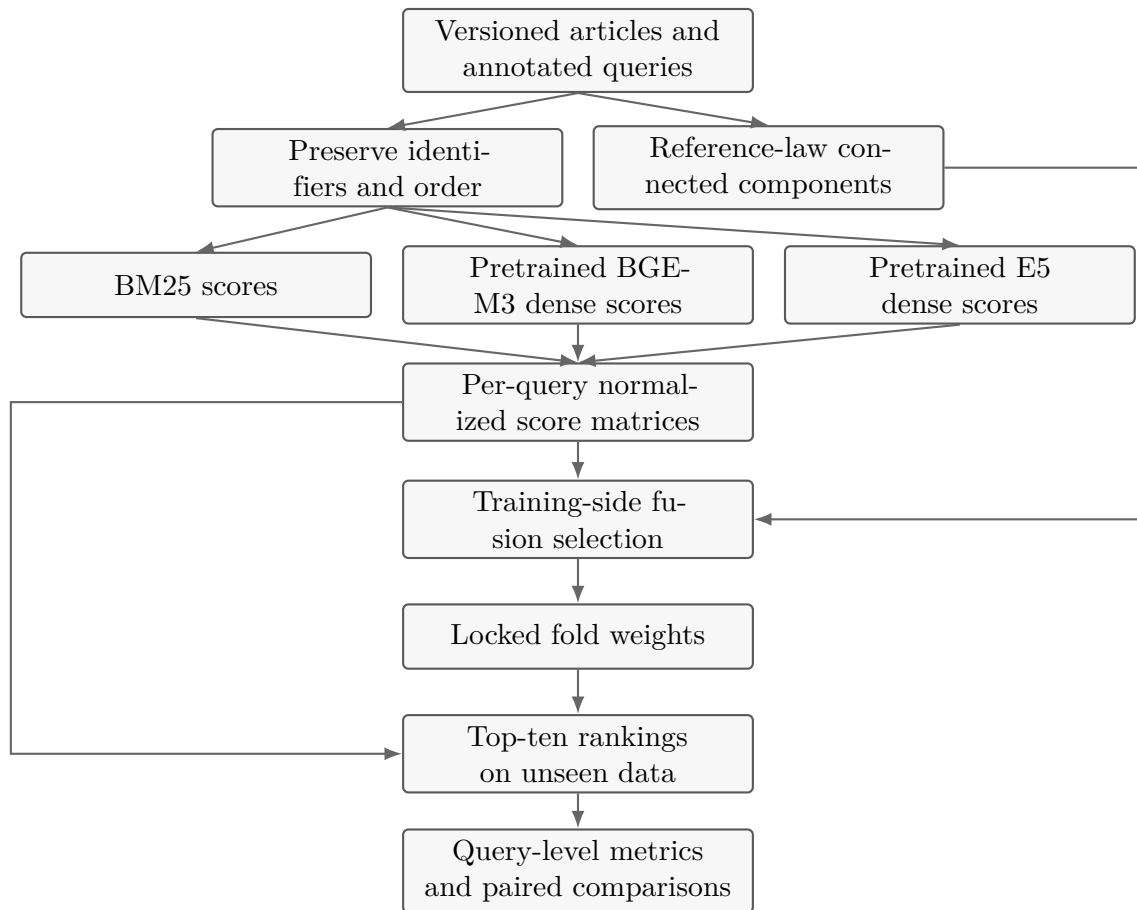


Figure 4.1: Baseline scoring and evaluation architecture.

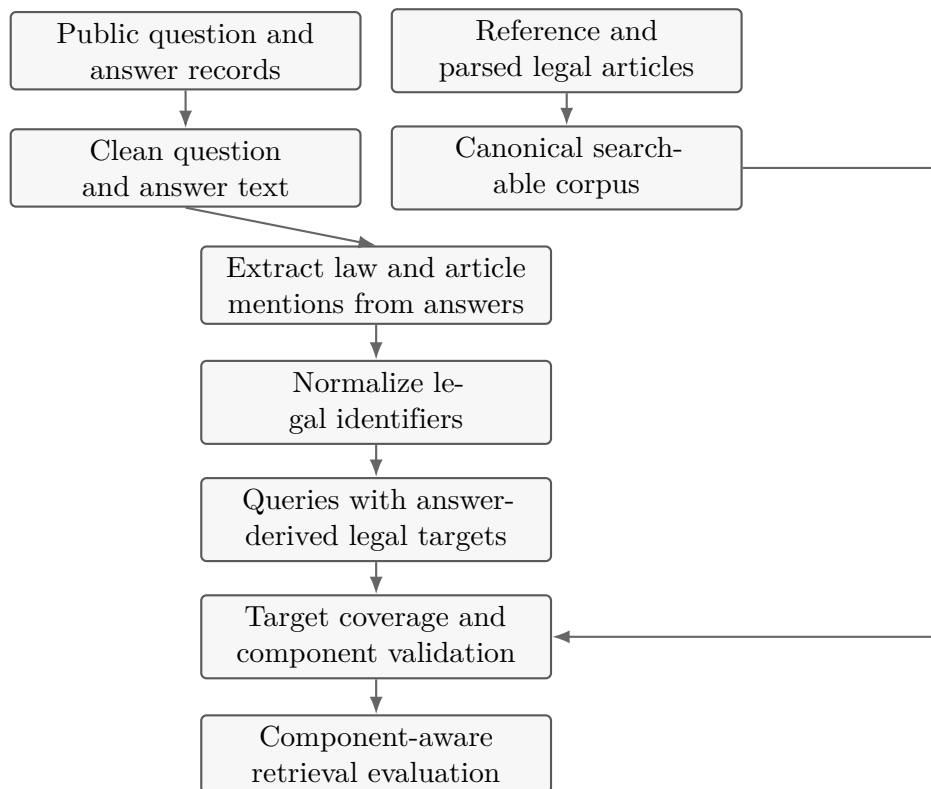


Figure 4.2: Construction of the BCA retrieval collection.

Table 4.2: Worked BCA question-to-target transformation.

Stage	Representation	Interpretation
Stored query	Vietnamese question about mandatory fire-prevention equipment for apartment buildings	Query text supplies the retrieval input
Extracted answer references	A cited decree with whole-law and article-level mentions	Published answer citations provide target candidates
Normalized law identity	Standardized law identifier for the cited decree	Citation text is aligned to the corpus identity space
Canonical target identity	Normalized law identifier plus article identifier	Evaluation matches ranked documents by legal identity
Retriever input	Query text and candidate article text	The published answer is not used as a query-time feature

The example demonstrates the collection design but not legal completeness. A published answer may omit useful supporting provisions, and an extracted citation may require legal review before being treated as an exhaustive relevance judgment.

4.4 Lexical and Dense Representations

BM25 scores are computed from tokenized query and article text. The lexical preprocessing lowercases text, removes non-word punctuation as token separators, and splits on whitespace. This simple policy is treated as part of the fixed lexical baseline. The experiments do not claim to evaluate every possible Vietnamese lexical analyzer or tokenization strategy. Dense scores are produced by pretrained BGE-M3 and multilingual-E5-base encoders. Query and article embeddings are L2-normalized, and dense similarity is the vector dot product:

$$s_{\text{dense}}(q, d) = \frac{\mathbf{u}_q^\top \mathbf{v}_d}{\max(\|\mathbf{u}_q\|_2, 10^{-10}) \max(\|\mathbf{v}_d\|_2, 10^{-10})}.$$

Table 4.3: Baseline dense-model identity.

Property	BGE-M3	Multilingual-E5-base
Model family	BGE-M3 multilingual embedding model	E5 multilingual text-embedding model
Embedding dimension	1,024	768
Experimental role	Primary pretrained dense component	Additional pretrained dense component
Baseline maximum input length	512 tokens	512 tokens
Adaptation status	Adapted in the fine-tuning studies	Frozen throughout the thesis

The evaluation uses exhaustive corpus scoring rather than an approximate nearest-neighbor index. Therefore, a mapped and available target outside the top ten reflects scoring and fusion behavior, not candidate-generation recall. This design simplifies scientific interpretation, although it does not establish production latency.

4.5 Score Normalization and Fusion

BM25 and dense scores have different numeric scales, so they are normalized before fusion. For retriever r , raw score $s_r(q, d)$, and corpus C , the implemented per-query min-max normalization is

$$\tilde{s}_r(q, d) = \frac{s_r(q, d) - \min_{d' \in C} s_r(q, d')}{\max(\max_{d' \in C} s_r(q, d') - \min_{d' \in C} s_r(q, d'), \epsilon)}.$$

The small constant ϵ prevents division by zero. Normalization is applied independently for each query and channel over the full corpus.

The two-way hybrid score is

$$S_2(q, d) = w_b \tilde{s}_b(q, d) + (1 - w_b) \tilde{s}_g(q, d),$$

where b denotes BM25 and g denotes BGE-M3. The three-way score is

$$S_3(q, d) = w_b \tilde{s}_b(q, d) + w_g \tilde{s}_g(q, d) + w_e \tilde{s}_e(q, d), \quad w_b + w_g + w_e = 1,$$

where e denotes E5. Weights are nonnegative. The evaluated grid uses increments of 0.1. Thus, the two-way grid contains 11 candidate settings and the three-way simplex contains

66 candidate settings, including zero-weight boundaries.

For each split, training-side mean NDCG@10 selects the fusion weights, which are then fixed for evaluation on unseen data. Because the three-way grid contains the two-way boundary, a higher training-side fit does not by itself prove a benefit on unseen data. The two-way versus three-way comparison on unseen data in Chapter 5 is therefore the relevant evidence for E5’s incremental value.

4.6 Evaluation Design

The baseline evaluation uses law-component grouping to reduce leakage from shared reference laws. Query groups are assigned to folds so that reference-law components are not split between training-side fusion selection and evaluation on unseen data. The resulting out-of-fold predictions measure a selection procedure rather than one universal global weight vector.

Algorithm 4.1 Grouped selection of fusion weights and out-of-fold evaluation

Require: Ordered queries Q , corpus D , aligned score matrices, grouped folds, candidate weight grid W

Ensure: Selected weights per fold and query metrics on unseen data

```

1: for each fold  $f$  do
2:   Let  $T$  be the training-side query indices and  $V$  the unseen-data indices
3:   for each candidate weight vector  $w \in W$  do
4:     Rank  $D$  for every query in  $T$  using the weighted score sum
5:     Compute mean query NDCG@10
6:   end for
7:   Select the first candidate with the highest training-side mean
8:   Rank all queries in  $V$  using the selected weights
9:   Store query-level metrics on unseen data
10: end for
11: Aggregate unseen-data query rows across folds

```

Algorithm 4.2 Article ranking with a locked hybrid configuration

Require: Query q , corpus D , retriever scores, locked fusion weights w

Ensure: Ten highest-ranked articles

```

1: Normalize each retriever’s scores over the full corpus for  $q$ 
2: for each article  $d \in D$  do
3:    $\text{combined}[d] \leftarrow \sum_r w_r \text{normalized\_score}[r, d]$ 
4: end for
5: return the ten articles with highest combined score

```

For BCA, ordinary query-level evaluation would overstate reliability because many questions are connected through shared legal targets. The corrected protocol distinguishes a giant-component holdout, small-component out-of-fold evaluation, and a combined descriptive view. Chapter 5 reports all three because they answer different questions: performance on the dominant connected component, performance across smaller components, and descriptive behavior over the entire retained BCA collection.

4.7 Legal-Domain Fine-Tuning

Fine-tuning tests whether legal-domain supervision improves BGE-M3. The adaptation studies change only the BGE-M3 dense component. BM25 and E5 remain fixed. The primary hybrid evaluation then replaces pretrained BGE-M3 with the adapted BGE-M3 while keeping the relevant fusion policy explicit. This thesis refers to the two experiments as Study A: Hybrid-Oriented Adaptation and Study B: Dense-Oriented Adaptation.

4.7.1 Training Pairs and Negatives

Study A: Hybrid-Oriented Adaptation uses Zalo query–article pairs. For each eligible query, the training builder selects a known relevant article available in both clean and parsed views, creating paired training examples from the two corpus representations. Negative passages are mined from lexical and dense rankings after excluding known positives, duplicates, unavailable documents, and laws reserved for evaluation on **unseen data**.

Table 4.4: Training groups and protected folds in the completed hybrid fine-tuning study.

Stage	Train folds	Queries	A/B groups	Protected folds	Protected laws	Law overlap
Pilot	2, 3, 4	1,917	3,834	0, 1	258	0
Final training	1, 2, 3, 4	2,556	5,112	0	110	0

The zero-overlap checks protect the law identifiers reserved for evaluation on unseen data from appearing in the constructed training groups. They do not prove absence of shared legal concepts or similar language, and they do not make the final evaluation a test on unseen legal systems.

4.7.2 Training Objective and Configuration

Study A trains BGE-M3 with a single-positive in-batch contrastive objective. Each query group contains one selected positive and seven explicit negatives. With batch size B , group size $G = 8$, positive passage index $y_i = iG$, normalized query vector $\mathbf{q}_i$, normalized passage vector $\mathbf{p}_j$, and temperature $\tau = 0.02$, the implemented objective is

$$\mathcal{L}_{\text{batch}} = -\frac{1}{B} \sum_{i=0}^{B-1} \log \frac{\exp(\mathbf{q}_i^\top \mathbf{p}_{y_i} / \tau)}{\sum_{j=0}^{BG-1} \exp(\mathbf{q}_i^\top \mathbf{p}_j / \tau)}.$$

Table 4.5: Locked configuration for the completed final-training runs.

Parameter	Verified value
Base encoder	BGE-M3
Adaptation	Full dense encoder
Epochs	2
Learning rate	2.0000×10^{-5}
Contrastive query batch	4
Gradient accumulation	8
Optimizer effective batch	32 query groups
Passage group	1 selected positive and 7 explicit negatives
Query / passage maximum length	256 / 512 tokens
Temperature	0.0200
Pooling / normalization	First-token hidden state / L2 normalization
Seeds	42, 43, 44
Frozen E5 component	Multilingual-E5-base, unchanged throughout adaptation

Pilot training uses folds 2–4; fold 1 selects the learning rate, epoch, and contrastive batch setting by hybrid NDCG@10. After the recipe is locked, final models are retrained on folds 1–4 using three seeds. Fold 0 is reserved for Study A’s corrected hybrid comparison on data unseen during training and selection. Because a flawed preliminary analysis had already exposed fold 0, the corrected result is treated as a post-hoc evaluation on unseen data, not as a pristine first-use test.

Figure 4.3 summarizes the selection and evaluation sequence.

4.7.3 Study B: Dense-Oriented Adaptation

Study B evaluates BGE-M3 as a dense retriever before integrating it into hybrids. This study uses a separate **recipe** with an all-positive contrastive mask, five explicit negatives, a larger true contrastive batch requirement, and a Zalo-only first passing data stage. Its selected setting is evaluated with three final seeds on the same 640 official queries used for the dense and fusion comparisons.

For logits $z_{ij} = s(q_i, d_j)/\tau$, positive candidate positions P_i , candidate positions C_i , and

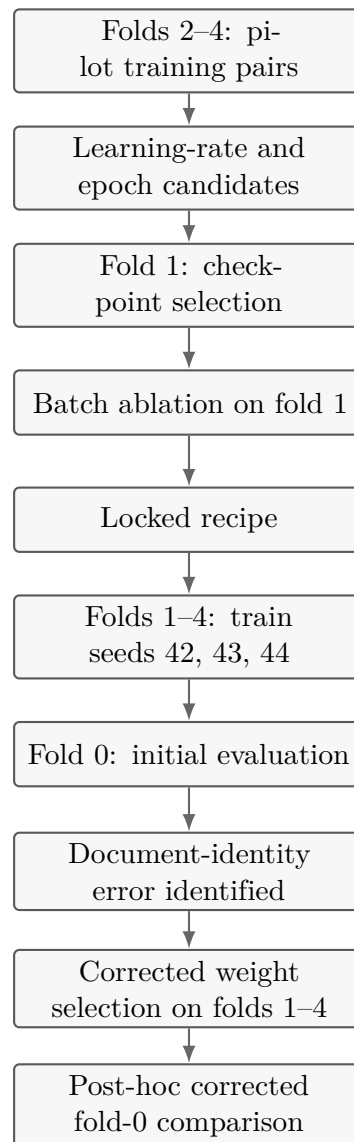


Figure 4.3: Training and evaluation evidence flow.

batch size B , the multi-positive loss is

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \left[\log \sum_{j \in P_i} \exp(z_{ij}) - \log \sum_{j \in C_i} \exp(z_{ij}) \right].$$

Study B is reported separately from Study A because it uses a different recipe and answers a different question: whether dense-only gains transfer into complete hybrids.

4.8 Method Summary

The methodology fixes the retrieval task at article ranking, preserves query and corpus identities, compares lexical and dense signals under selected fusion weights, and evaluates legal-domain adaptation only after explicit selection boundaries. BCA adds a dataset-construction and evaluation-design contribution by exposing coverage and component-dependence limits. Chapter 5 reports the resulting effectiveness, failure analysis, and adaptation outcomes.

Chapter 5

Experimental Evaluation

This chapter reports the experiments in the order of the research questions. It first evaluates lexical and hybrid retrieval, then isolates the incremental contribution of E5, then examines BCA coverage and component-aware reliability, and finally analyzes legal-domain adaptation. The central adaptation finding is that dense retriever improvement does not necessarily transfer into a complete hybrid retrieval system.

5.1 Datasets and Experimental Setup

Quantitative table values are displayed to four decimal places, using scientific notation with four fractional digits in the mantissa for tiny nonzero values. Deltas are calculated before display rounding; integer counts and indices remain integers.

The evaluated conditions and corpus sizes are defined in Table 4.1. ALQAC and Zalo each have clean and parsed corpus representations; their A/B conditions share query identities and are therefore paired corpus conditions rather than independent datasets. **BCA is evaluated separately because many questions share legal targets and because some targets are unavailable in the indexed corpus.**

NDCG@10 is the primary endpoint. Hit@10, MRR@10, and MAP@10 are reported to distinguish top-ten recovery, first-relevant ordering, and multiple-target ranking. ALQAC and Zalo use query-level paired comparisons for the two-way versus three-way hybrid. BCA uses component-aware analysis in addition to descriptive query-level summaries. These units of inference are not interchangeable.

Table 5.1: System identities and comparison boundaries.

Configuration	Baseline role	Adaptation role
BM25	Fixed lexical reference	Unchanged lexical component
BGE-M3 dense	Pretrained dense retriever	Adapted in the legal-domain fine-tuning studies
E5 dense	Additional pretrained dense retriever	Frozen throughout the thesis

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Configuration	Baseline role	Adaptation role
BM25+BGE-M3	Two-way lexical–semantic hybrid	Evaluated with pretrained and adapted BGE-M3
BM25+ BGE-M3+E5	Three-way hybrid	Evaluated with pretrained and adapted BGE-M3 while E5 remains fixed

The experiments compare methods within their own query populations and selection boundaries. Results from different folds, corpora, or **adaptation recipes** are not pooled into a single cross-protocol score.

5.2 Lexical and Hybrid Retrieval

This section compares lexical–semantic hybrid retrieval with the **fixed BM25 reference**. Table 5.2 reports BM25, pretrained dense retrievers, and hybrid systems for ALQAC and Zalo. Table 5.3 summarizes the hybrid changes relative to BM25 and includes BCA’s combined descriptive view.

Table 5.2: Pretrained standalone and hybrid results, query-macro OOF population.

Dataset	Condition	Method	NDCG@10	MRR@10	MAP@10	Hit@10
ALQAC	A	BM25	0.7986	0.7648	0.7621	0.9112
	A	BGE-M3 pretrained	0.8437	0.8134	0.8108	0.9433
	A	E5 pretrained	0.8126	0.7729	0.7722	0.9371
	A	BM25+BGE-M3	0.8805	0.8534	0.8516	0.9692
	A	BM25+BGE-M3+E5	0.8784	0.8519	0.8504	0.9630
	B	BM25	0.6479	0.6175	0.6159	0.7472
	B	BGE-M3 pretrained	0.6684	0.6401	0.6382	0.7620
	B	E5 pretrained	0.6510	0.6177	0.6165	0.7583
	B	BM25+BGE-M3	0.7067	0.6812	0.6797	0.7904
	B	BM25+BGE-M3+E5	0.7055	0.6805	0.6793	0.7867
Zalo	A	BM25	0.5264	0.4731	0.4682	0.7100
	A	BGE-M3 pretrained	0.6424	0.5917	0.5850	0.8235

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Condition	Method	NDCG@10	MRR@10	MAP@10	Hit@10
Zalo A	E5 pretrained	0.5640	0.5104	0.5046	0.7509
Zalo A	BM25+BGE-M3	0.6805	0.6298	0.6235	0.8583
Zalo A	BM25+BGE-M3+E5	0.6818	0.6317	0.6254	0.8576
Zalo B	BM25	0.5620	0.5099	0.5050	0.7406
Zalo B	BGE-M3 pretrained	0.6889	0.6431	0.6366	0.8526
Zalo B	E5 pretrained	0.6259	0.5748	0.5689	0.8054
Zalo B	BM25+BGE-M3	0.7214	0.6767	0.6705	0.8789
Zalo B	BM25+BGE-M3+E5	0.7344	0.6913	0.6845	0.8889

BGE-M3 is the strongest individual retriever by NDCG@10 in all four ALQAC/Zalo conditions. Both hybrids exceed BM25, and both also exceed the strongest individual component in these conditions. The result supports lexical-semantic complementarity at the system level, although it does not identify the linguistic mechanism behind each gain.

Table 5.3: Hybrid NDCG@10 changes relative to fixed BM25.

Condition	Hybrid	BM25	Hybrid score	Absolute change	Relative change
nên in đậm cái nào tốt nhất					
ALQAC A	BM25+BGE-M3	0.7986	0.8805	+0.0819	+10.2590%
ALQAC A	BM25+BGE-M3+E5	0.7986	0.8784	+0.0798	+9.9970%
ALQAC B	BM25+BGE-M3	0.6479	0.7067	+0.0588	+9.0690%
ALQAC B	BM25+BGE-M3+E5	0.6479	0.7055	+0.0576	+8.8910%
Zalo A	BM25+BGE-M3	0.5264	0.6805	+0.1541	+29.2760%
Zalo A	BM25+BGE-M3+E5	0.5264	0.6818	+0.1554	+29.5210%
Zalo B	BM25+BGE-M3	0.5620	0.7214	+0.1594	+28.3680%
Zalo B	BM25+BGE-M3+E5	0.5620	0.7344	+0.1724	+30.6830%
BCA combined descriptive	BM25+BGE-M3	0.1687	0.2283	+0.0597	+35.3820%

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Condition	Hybrid	BM25	Hybrid score	Absolute change	Relative change
BCA combined descriptive	BM25+ BGE-M3+E5	0.1687	0.2297	+0.0610	+36.1760%

The hybrid gains over BM25 are large in Zalo and smaller but still positive in ALQAC. BCA also shows a positive descriptive difference from BM25, but its lower absolute scores must be interpreted with coverage limits and component dependence. The answer to RQ1 is therefore positive within the evaluated conditions: lexical-semantic hybrids improve the fixed BM25 reference. The result should not be generalized into a claim that one hybrid configuration will transfer unchanged to every legal corpus.

5.3 Incremental Contribution of E5

This section tests whether E5 adds value beyond BM25+BGE-M3. This is a stricter question than asking whether E5 is a competent standalone retriever. The relevant comparison is the two-way hybrid against the three-way hybrid.

Table 5.4: Incremental E5 contribution on the canonical ALQAC and Zalo populations.

Condition	Two-way NDCG	Three-way NDCG	Absolute change	Relative change	Stored Wilcoxon p
ALQAC A	0.8805	0.8784	-0.0021	-0.2380%	0.4498
ALQAC B	0.7067	0.7055	-0.0012	-0.1630%	0.7449
Zalo A	0.6805	0.6818	+0.0013	+0.1900%	0.1548
Zalo B	0.7214	0.7344	+0.0130	+1.8030%	5.1598×10^{-7}

bỏ two way trong bảng, thay bằng bm25+các kiểu

The incremental E5 result is condition-dependent. ALQAC shows small negative NDCG@10 differences in both corpus representations. Zalo A shows a very small positive difference, while Zalo B shows the clearest positive result: an absolute NDCG@10 gain of approximately 0.0130, with higher MRR@10 and Hit@10 as well. The p-values in Table 5.4 are reported as stored query-level paired tests; they do not remove the need to interpret multiple conditions and possible query dependence carefully.

The A/B corpus comparison also resists a simple parser narrative. ALQAC has substantially lower hybrid scores in the parsed condition than in the clean condition. Zalo, in contrast, has higher scores in the parsed-with-fallback condition than in the clean condition. This does not imply that parsing is inherently harmful for ALQAC and inherently beneficial for Zalo. The two collections differ in construction, target mapping, fallback behavior, and

article universe. A matched no-fallback ablation was not available, so the independent effect of fallback remains unmeasured.

The answer to RQ2 is therefore not a general yes. E5 provides useful incremental value in the parsed Zalo condition, negligible or unsupported value in clean Zalo, and no observed improvement in ALQAC. This is one of the thesis’s main practical findings: adding a dense retriever expands the search space, but it does not guarantee better ranking on unseen data.

5.4 BCA Coverage and Component-Aware Evaluation

BCA serves two purposes. It extends the evaluation beyond ALQAC and Zalo, and it exposes how corpus coverage and target dependence affect interpretation. Table 5.5 reports BCA results under the giant-component, small-component, and combined descriptive views.

Table 5.5: Compact BCA protocol and coverage views. All metrics are at cutoff 10; OOF denotes out-of-fold.

Protocol	Cohort	Method	N	NDCG	MRR	MAP	Hit
Giant	All	BM25	484	0.1670	0.2469	0.1098	0.4545
Giant	All	BGE-M3	484	0.2153	0.3143	0.1453	0.5455
Giant	All	BM25+BGE	484	0.2275	0.3414	0.1573	0.5517
Giant	All	BM25+BGE+E5	484	0.2295	0.3413	0.1560	0.5702
Giant	Answerable	BM25	394	0.2051	0.3033	0.1348	0.5584
Giant	Answerable	BGE-M3	394	0.2645	0.3861	0.1785	0.6701
Giant	Answerable	BM25+BGE	394	0.2795	0.4194	0.1932	0.6777
Giant	Answerable	BM25+BGE+E5	394	0.2819	0.4193	0.1916	0.7005
Small OOF	All	BM25	62	0.1817	0.2157	0.1382	0.3387
Small OOF	All	BGE-M3	62	0.2015	0.2392	0.1597	0.3387
Small OOF	All	BM25+BGE	62	0.2346	0.2921	0.1865	0.3871
Small OOF	All	BM25+BGE+E5	62	0.2309	0.2838	0.1815	0.3871
Small OOF	Answerable	BM25	46	0.2449	0.2907	0.1863	0.4565
Small OOF	Answerable	BGE-M3	46	0.2715	0.3225	0.2153	0.4565
Small OOF	Answerable	BM25+BGE	46	0.3162	0.3937	0.2514	0.5217
Small OOF	Answerable	BM25+BGE+E5	46	0.3112	0.3825	0.2446	0.5217
Combined OOF	All	BM25	546	0.1687	0.2434	0.1130	0.4414
Combined OOF	All	BGE-M3	546	0.2137	0.3058	0.1469	0.5220
Combined OOF	All	BM25+BGE	546	0.2283	0.3358	0.1606	0.5330
Combined OOF	All	BM25+BGE+E5	546	0.2297	0.3348	0.1589	0.5495
Combined OOF	Answerable	BM25	440	0.2093	0.3020	0.1402	0.5477
Combined OOF	Answerable	BGE-M3	440	0.2652	0.3795	0.1823	0.6477
Combined OOF	Answerable	BM25+BGE	440	0.2833	0.4167	0.1993	0.6614
Combined OOF	Answerable	BM25+BGE+E5	440	0.2850	0.4154	0.1972	0.6818

The BCA reference graph contains one giant component with 484 of 546 queries and 49 smaller components containing the remaining 62 queries. This structure affects statistical

interpretation. The combined descriptive view shows the three-way hybrid slightly above the two-way hybrid in NDCG@10, but the small-component out-of-fold view shows a negative NDCG@10 difference for the three-way system. The corresponding component-aware inference does not establish an E5 advantage.

Coverage is also a major limitation. Of the 546 BCA queries, 440 have at least one represented target and 106 have none. A further 229 answerable queries are partial-target cases in which some, but not all, targets are present. An unanswerable query cannot receive a positive Hit@10 from any ranking method over the fixed corpus. Therefore, BCA’s lower end-to-end scores partly reflect corpus and mapping limitations rather than only scoring failure.

The BCA result is a dataset and evaluation contribution rather than a simple model leaderboard. It shows that an end-to-end legal retrieval score can mix three phenomena: whether the evidence exists, whether it is mapped to the correct legal identity, and whether the retriever ranks it highly. A method improvement can address only the third problem unless corpus construction changes as well.

5.5 Failure Analysis

The failure analysis asks what a model intervention could plausibly repair. Missing targets require corpus or mapping work. Available targets outside the top ten require ranking or fusion improvements. Relevant articles demoted within the top ten affect NDCG and MRR even when Hit@10 is unchanged. These distinctions guide the adaptation experiment but do not prove that fine-tuning will solve the observed failures.

BCA provides the clearest evidence of non-rankable failures: 106 questions have no mapped target in the corpus. For these questions, changing BM25 weights, dense encoders, or fusion rules cannot recover the target. Partial-target queries create a second limitation: a system may retrieve an available relevant article while still being penalized for missing other targets that are absent from the corpus.

For ALQAC and Zalo, top-ten boundary cases show that E5 can both recover and lose hits. The same added signal that improves parsed Zalo on average can demote specific relevant articles below the cutoff. Appendix E preserves representative cases. The main conclusion is more general: aggregate improvement does not imply uniform query-level improvement, and aggregate regression does not imply that every query becomes worse.

The analysis remains an information-retrieval analysis. It identifies target absence, top-ten transitions, within-law article demotion, and persistent misses. It does not assign legal-reasoning causes such as misunderstanding exceptions or temporal validity. A legally adjudicated error taxonomy would require a sampling plan, full text review, annotation rules, and reviewer agreement. That study was not performed here.

5.6 Legal-Domain Adaptation

This section examines whether legal-domain BGE-M3 adaptation improves retrieval beyond development selection. The narrative has six parts: Study A development selection, Study A corrected hybrid evaluation on unseen data, Study B dense-only adaptation, transfer of dense improvements into hybrid retrieval, query-level analysis, and overall interpretation.

5.6.1 Development Selection

Study A: Hybrid-Oriented Adaptation selects a BGE-M3 training recipe on development queries. Table 5.6 reports the learning-rate and epoch pilot under passage length 512. Table 5.7 reports the subsequent batch ablation at the selected learning rate and epoch.

Table 5.6: Passage-512 learning-rate and epoch pilot: development NDCG@10.

Learning rate	Epoch	Two-way hybrid	Three-way hybrid
5.0000×10^{-6}	1	0.7470	0.7685
5.0000×10^{-6}	2	0.7603	0.7785
5.0000×10^{-6}	3	0.7647	0.7815
1.0000×10^{-5}	1	0.7509	0.7816
1.0000×10^{-5}	2	0.7658	0.7873
1.0000×10^{-5}	3	0.7628	0.7897
2.0000×10^{-5}	1	0.7440	0.7757
2.0000×10^{-5}	2	0.7691	0.7923
2.0000×10^{-5}	3	0.7689	0.7958

The pilot shows development improvements across several candidate settings. The selected recipe uses learning rate 0.00002 and two epochs. The choice is not evidence that the selected point is a global optimum; it is the best supported point under the restricted search and selection rule.

Table 5.7: Batch ablation at the selected learning rate and epoch; scores are development NDCG@10.

Batch/accum.	Two-way	Δ	Three-way	Δ
Pretrained baseline	0.7172	–	0.7245	–
1 / 32	0.7725	+0.0553	0.7869	+0.0624
2 / 16	0.7693	+0.0521	0.7924	+0.0679
4 / 8	0.7781	+0.0609	0.7934	+0.0689

The selected development configuration improves two-way NDCG@10 by 0.060945 and three-way NDCG@10 by 0.068882. This justifies testing the intervention on unseen data. It does not establish generalization, because the development queries are part of model selection.

5.6.2 Corrected Hybrid Evaluation on Unseen Data

The corrected comparison on unseen data evaluates Zalo parsed fold 0 with 640 queries. Fusion weights are fitted on the permitted training-side folds using pretrained BGE-M3 and then held fixed when BGE-M3 is replaced by its adapted version. This design measures encoder replacement within a specified hybrid rather than unrestricted retuning of the whole system.

The corrected two-way weights are $\text{BM25/BGE/E5} = 0.2/0.8/0.0$. The corrected three-way weights are $0.2/0.5/0.3$. Because a flawed preliminary analysis had already exposed the same fold, the corrected comparison is interpreted as a post-hoc evaluation on data unseen during training and selection, not as a pristine first-use test.

Table 5.8: Corrected Zalo fold-0 two-way hybrid evaluation; all metrics are at cutoff 10.

BGE state	Seed	NDCG	Δ	MRR	MAP	Hit
Pretrained	Fixed	0.7516	–	0.7015	0.6985	0.9141
Fine-tuned	42	0.7005	–0.0511	0.6470	0.6451	0.8750
Fine-tuned	43	0.7005	–0.0511	0.6490	0.6467	0.8688
Fine-tuned	44	0.6977	–0.0539	0.6398	0.6374	0.8859
Fine-tuned	Mean	0.6995	–0.0520	0.6453	0.6430	0.8766

All three two-way seeds score below the pretrained two-way baseline. The mean NDCG@10 change is -0.052045, and MRR@10, MAP@10, and Hit@10 are also lower. The result therefore cannot be rescued by choosing a different seed from the three reported runs.

Table 5.9: Corrected Zalo fold-0 three-way hybrid evaluation; all metrics are at cutoff 10.

BGE state	Seed	NDCG	Δ	MRR	MAP	Hit
Pretrained	Fixed	0.7595	–	0.7077	0.7048	0.9266
Fine-tuned	42	0.7320	–0.0275	0.6771	0.6743	0.9141
Fine-tuned	43	0.7340	–0.0255	0.6771	0.6747	0.9203
Fine-tuned	44	0.7378	–0.0217	0.6794	0.6771	0.9281
Fine-tuned	Mean	0.7346	–0.0249	0.6779	0.6754	0.9208

The three-way hybrid also has lower mean NDCG@10 after fine-tuning. The regression is smaller than in the two-way system, but it remains a regression against the matched pretrained three-way comparator. Seed 44 slightly improves Hit@10 while reducing

NDCG@10 and MRR@10, showing why the primary rank-sensitive metric must remain central.

Table 5.10: Corrected paired statistics for NDCG@10.

Comparison	Queries	Clusters	Mean Δ	95% cluster CI	Raw p	Holm p
Fine-tuned minus pretrained, two-way	640	106	-0.0520	$[-0.0919, +0.1076]$	0.7480	1.0000
Fine-tuned minus pretrained, three-way	640	106	-0.0249	$[-0.0542, +0.0938]$	0.7480	1.0000

The intervals in Table 5.10 include zero, so the observed regressions are not described as statistically established population-level harm. However, non-significance is not equivalence, and the practical improvement threshold of +0.01 NDCG@10 is not met. The correct conclusion is that the selected adaptation recipe failed to demonstrate hybrid improvement on unseen data under this protocol.

5.6.3 Study B: Dense-Oriented Adaptation

Study A does not by itself determine whether adaptation failed at the dense-retriever level or only after integration into the hybrid. Study B directly evaluates dense-only BGE-M3 adaptation before testing hybrid transfer.

Table 5.11: Study B post-hoc dense benchmark on 640 official queries.

System	NDCG@10	Delta NDCG@10	MRR@10	Recall@100
Pretrained	0.6740	0.0000	0.6213	0.9617
Fine-tuned seed 42	0.6932	0.0192	0.6448	0.9531
Fine-tuned seed 43	0.6925	0.0185	0.6431	0.9516
Fine-tuned seed 44	0.6928	0.0189	0.6437	0.9531
Fine-tuned seed mean	0.6928	0.0189	0.6438	0.9526

Study B shows a mean NDCG@10 gain of 0.018866, with all three seeds improving this endpoint. Mean Recall@100 decreases by 0.009115. The dense result therefore supports the claim that legal-domain fine-tuning can improve the dense encoder on this fixed-corpus query benchmark, while also showing a small recall tradeoff at depth 100.

5.6.4 Transfer of Dense Improvements into Hybrid Retrieval

The crucial question is whether the dense improvement remains useful inside the full hybrid. Table 5.12 reports the Study B primary locked-weight hybrid outcomes. Fusion

weights are selected with pretrained BGE-M3 and then held fixed for the corresponding fine-tuned comparison.

Table 5.12: Study B primary fusion outcomes under pretrained-locked weights.

Hybrid	Pretrained	FT mean	Δ NDCG	Rel. Δ	Δ R@100	Positive seeds	Gate
Two-way	0.7526	0.7483	-0.0043	-0.5707%	-0.0057	0/3	Fail
Three-way	0.7467	0.7476	0.0009	0.1203%	-0.0031	2/3	Fail

The dense-only gain does not transfer into a practically useful hybrid gain. The two-way hybrid decreases by 0.004295 NDCG@10. The three-way hybrid increases by only 0.000898 NDCG@10, far below the +0.01 improvement threshold. Thus, Study B sharpens the main RQ3 finding: adaptation can improve standalone dense ranking while failing to improve the complete hybrid retrieval system by a meaningful amount.

The mechanism is not isolated by these experiments. Possible explanations include changed score calibration, overlap between lexical and dense evidence, and fusion weights selected for pretrained scores. Because these mechanisms are not independently varied, they remain hypotheses for future work rather than established causes.

5.6.5 Query-Level Analysis

Aggregate means can hide mixtures of repaired and introduced failures. Table 5.13 summarizes the Study A corrected seed-42 hybrid changes. The first three rows partition all 640 queries by NDCG@10 change; the final rows describe top-ten transitions and are not additional disjoint totals.

Table 5.13: Query-level changes after fine-tuning, corrected seed 42.

Outcome	Two-way hybrid	Three-way hybrid
Higher NDCG@10	102	102
Lower NDCG@10	210	163
Unchanged NDCG@10	328	375
Previously missing at ten, now retrieved	23	16
Previously retrieved at ten, now missing	48	24
Missing at ten in both systems	32	31

The two-way system repairs some top-ten misses, but introduces more than twice as many new top-ten misses as it repairs. The three-way system introduces fewer new misses, consistent with its smaller aggregate regression, but it still introduces more misses than it

repairs. These counts show why the negative mean should not be interpreted as uniform deterioration, and why isolated repaired examples cannot overturn the aggregate result.

Table 5.14: Study B NDCG changes by query, using mean fine-tuned seed metric minus pretrained.

Comparison	Improved	Worse	Tied
Dense	127	96	417
Two-way	81	93	466
Three-way	85	86	469

Study B’s dense-only comparison has more improved than worsened queries, while the Study B hybrids do not show the same favorable balance. This query-level view is consistent with the aggregate finding: dense improvement exists, but much of it is absorbed or neutralized when the adapted encoder is placed inside the full fusion system.

5.6.6 Overall Interpretation

The RQ3 evidence has a clear sequence. First, Study A’s development selection shows large gains and justifies testing the adaptation. Second, Study A’s corrected hybrid evaluation on unseen data shows regressions for both two-way and three-way hybrids. Third, Study B shows that legal fine-tuning can improve dense BGE-M3 ranking. Fourth, that dense gain largely fails to transfer into locked-weight hybrid retrieval.

This sequence supports a stronger and more nuanced conclusion than either “fine-tuning works” or “fine-tuning fails.” It works at the dense-retriever level in Study B, but it does not necessarily improve the complete lexical-semantic system. The complete system must therefore be evaluated directly. Development scores and dense-only scores are insufficient substitutes for end-to-end hybrid evaluation.

Several boundaries remain. Cross-collection adaptation to ALQAC and BCA was not established. Secondary retuned-weight hybrid results were not available, so the thesis does not claim that the adapted encoder could not be made useful under a different fusion policy. A fresh untouched cohort would be required for a cleaner confirmatory test. These limitations narrow the conclusion without weakening the observed negative transfer result.

5.7 Experimental Synthesis

The experiments answer the research questions as follows. For RQ1, hybrid retrieval improves the fixed BM25 reference in the evaluated ALQAC, Zalo, and descriptive BCA conditions. For RQ2, E5 has condition-dependent incremental value: parsed Zalo provides

the clearest positive case, while ALQAC and BCA do not support a general three-way advantage. For RQ3, legal-domain adaptation improves development scores and can improve dense-only ranking, but it does not necessarily improve the complete hybrid retrieval system.

The BCA analysis adds a methodological lesson. Evaluation cannot be reduced to a ranking table when target coverage and query dependence are substantial. Missing targets, partial target availability, and giant connected components affect what a retrieval score means. Reporting both end-to-end and answerable-only views makes this limitation visible rather than hiding it inside an average.

The principal scientific takeaway is that legal retrieval systems must be evaluated at the level at which they will be used. A model can be better in isolation while adding little to a hybrid; an added retriever can help one corpus representation and not another; a retrieval failure can be caused by absent evidence rather than poor ranking. These findings justify the thesis's emphasis on controlled comparison, negative results, and careful limitation of causal claims.

Chapter 6

Conclusion and Future Work

This chapter synthesizes the experimental findings rather than repeating the numerical analysis from Chapter 5. The conclusions distinguish improvement over BM25, incremental value beyond an existing hybrid, and transfer from dense adaptation into a complete retrieval system.

6.1 Summary of Findings

6.1.1 Hybrid Retrieval over the Lexical Baseline

The experiments support the value of lexical–semantic hybrid retrieval over the fixed BM25 reference in the evaluated conditions. Both the BM25+BGE-M3 hybrid and the BM25+BGE-M3+E5 hybrid improve NDCG@10 over BM25 across the reported ALQAC and Zalo conditions, and BCA shows the same direction in its combined descriptive view. The result is strongest as an empirical finding about these collections and corpus representations. It does not imply that a single weight setting, corpus construction policy, or hybrid architecture will transfer unchanged to other legal corpora.

6.1.2 Incremental Contribution of E5

The incremental value of E5 is condition-dependent. The clearest positive result occurs on parsed Zalo, where adding E5 increases NDCG@10 by approximately 0.0130. ALQAC shows small negative differences, clean Zalo shows only a small positive difference, and BCA does not establish a reliable small-component advantage. Therefore, the thesis does not support a universal claim that three-way retrieval is better than two-way retrieval. The value of the additional dense retriever must be evaluated against a strong two-way reference in the target corpus condition.

6.1.3 Effect of Legal-Domain Adaptation

The answer is mixed and is one of the main findings of the thesis. Study A: Hybrid-Oriented Adaptation shows substantial development-selection gains, but its corrected hybrid evaluation on data unseen during training and selection shows mean NDCG@10

decreases of 0.0520 for the two-way hybrid and 0.0249 for the three-way hybrid. Study B: Dense-Oriented Adaptation shows that fine-tuning can improve BGE-M3 as a dense retriever, increasing mean NDCG@10 by 0.0189. However, that improvement largely disappears after the adapted encoder is integrated into locked-weight hybrids. Thus, dense adaptation can improve an encoder without improving the complete lexical–semantic retrieval system by a meaningful amount.

6.2 Research Contributions

The first contribution is a controlled empirical evaluation of lexical–semantic hybrid retrieval for Vietnamese legal information retrieval. The thesis compares BM25, BGE-M3, E5, two-way fusion, and three-way fusion across multiple legal collections and corpus representations. It separates improvement over a lexical baseline from incremental improvement over an already strong hybrid.

The second contribution is an evaluation of legal-domain BGE-M3 adaptation at both dense-retriever and complete-hybrid levels. The work preserves negative and non-transfer results rather than treating them as failed experiments. The distinction between dense-only improvement and hybrid-system improvement is practically important for legal retrieval systems, where the deployed system is usually a pipeline rather than a single encoder.

The third contribution is a dataset and evaluation contribution involving the BCA collection. The thesis constructs a retrieval collection from public questions and published answer citations, analyzes target coverage, and applies component-aware evaluation. This contribution shows that corpus availability and query dependence can materially affect how legal retrieval metrics should be interpreted.

6.3 Discussion and Limitations

The experiments show that hybrid retrieval is useful, but they also show why retrieval evaluation should not stop at a favorable aggregate score. A hybrid can improve over BM25 while an additional component contributes little. A model can repair some queries while introducing other failures. A benchmark can contain many questions while having fewer independent legal components than the query count suggests. These distinctions matter in legal retrieval because a system must retrieve specific provisions, not merely topically related documents.

The E5 result illustrates the limits of model-count reasoning. Adding another dense retriever increases representational diversity and expands the fusion search space, but the measured benefit depends on the collection and corpus representation. Parsed Zalo

benefits most clearly; ALQAC does not. This finding supports local validation rather than a generic architectural rule.

The adaptation results illustrate the limits of development selection. Study A’s fine-tuning recipe performs well on development queries but regresses in corrected hybrid evaluation on unseen data. Study B shows that this is not simply because legal-domain fine-tuning is incapable of improving dense retrieval. Instead, the complete hybrid can absorb, dilute, or counteract dense improvements because fusion weights, lexical evidence, and score calibration all interact. The present experiments do not isolate which mechanism dominates.

The publicly available Vietnamese legal data suitable for supervised retrieval adaptation remain limited in the present setting. Additional diverse and carefully validated query–article training pairs may improve the fine-tuning process, particularly by providing broader coverage of legal topics and relevance relations. However, the study did not evaluate this intervention; it is therefore a hypothesis for future work rather than an empirical conclusion.

BCA exposes dataset limitations that are not model limitations. Some targets are absent from the indexed corpus, and some answerable queries are only partially covered. A ranking model cannot retrieve evidence that is not present. Therefore, end-to-end scores should be interpreted as measuring the whole retrieval pipeline, including corpus construction and target mapping, not only scoring quality.

Several validity limits remain. The evaluation is restricted to Vietnamese legal collections and recorded relevance judgments. It does not establish multilingual generalization, cross-jurisdiction transfer, downstream legal-answer correctness, current legal validity, user satisfaction, latency, throughput, or deployment cost. Study A’s corrected adaptation evaluation on data unseen during training and selection is post-hoc because it follows a flawed preliminary analysis of that same evaluation fold. Study B also uses a fixed-corpus benchmark with prior test observation, so it should not be described as a pristine prospective confirmation.

Statistical conclusions are also bounded. Non-significant intervals do not prove equivalence, and detectable differences do not automatically imply practical importance. The thesis therefore distinguishes observed mean changes, practical improvement thresholds, and statistical uncertainty.

6.4 Future Work

A new, prospectively controlled evaluation should be conducted on an untouched query set. Corpus construction, model selection, fusion configuration, and statistical procedures should be fixed before examining the final test results.

Corpus quality should also be improved, particularly for the BCA collection. Coverage gaps can be addressed through version-aware mapping of legal documents, while query-level failures can be systematically sampled and annotated using a defined rubric. This would help distinguish missing evidence, annotation limitations, lexical mismatch, semantic confusion, and fusion-related ranking errors.

Future work should further extend the evaluation from retrieval performance to the complete RAG system. This includes assessing answer correctness, citation support, legal sufficiency, latency, throughput, and computational cost. Retrieval metrics remain useful indicators, but they cannot replace direct evaluation of generated legal answers and deployed system performance.

6.5 Closing Remarks

This thesis shows that Vietnamese legal retrieval benefits from lexical–semantic hybridization, but also that more components or better dense-only scores do not automatically produce a better complete system. The most important lesson is methodological: legal retrieval claims must be tied to the exact corpus, target coverage, selection boundary, and system level being evaluated. A disciplined negative result can be as informative as a positive result when it prevents an unsupported deployment claim.

References

- [1] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, U.K.: Cambridge University Press, 2008. [Online]. Available: <https://nlp.stanford.edu/IR-book/>
- [2] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” in *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 2318–2335, 2024. [Online]. Available: <https://aclanthology.org/2024.findings-acl.137/>
- [3] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, and F. Wei, “Multilingual E5 text embeddings: A technical report,” arXiv:2402.05672, 2024. [Online]. Available: <https://arxiv.org/abs/2402.05672>
- [4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, 2020. [Online]. Available: <https://arxiv.org/abs/2005.11401>
- [5] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009. [Online]. Available: <https://www.nowpublishers.com/article/DownloadEBook/INR-019>
- [6] G. M. Rosa, R. C. Rodrigues, R. Lotufo, and R. Nogueira, “Yes, BM25 is a strong baseline for legal case retrieval,” in *COLIEE 2021*, 2021. [Online]. Available: <https://arxiv.org/abs/2105.05686>
- [7] R. Goebel, Y. Kano, M.-Y. Kim, J. Rabelo, K. Satoh, and M. Yoshioka, “Overview of benchmark datasets and methods for the Legal Information Extraction/Entailment Competition (COLIEE) 2024,” in *JSAI-isAI 2024*, LNAI 14741, pp. 109–124, 2024. [Online]. Available: https://coliee.org/documents/waivers/overview_COLIEE2024.pdf
- [8] ALQAC Organizers, *Automated Legal Question Answering Competition 2024*, 2024. [Online]. Available: <https://sites.google.com/view/alqac-2024/home>

- [9] D. Q. Nguyen and A. T. Nguyen, “PhoBERT: Pre-trained language models for Vietnamese,” in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020. [Online]. Available: <https://aclanthology.org/2020.findings-emnlp.92/>
- [10] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. EMNLP-IJCNLP 2019*, 2019. [Online]. Available: <https://aclanthology.org/D19-1410/>
- [11] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih, “Dense passage retrieval for open-domain question answering,” in *Proc. EMNLP 2020*, pp. 6769–6781, 2020. [Online]. Available: <https://aclanthology.org/2020.emnlp-main.550/>
- [12] O. Khattab and M. Zaharia, “ColBERT: Efficient and effective passage search via contextualized late interaction over BERT,” in *Proc. 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020. [Online]. Available: <https://arxiv.org/abs/2004.12832>
- [13] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, “Reciprocal rank fusion outperforms Condorcet and individual rank learning methods,” in *Proc. 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 758–759, 2009. [Online]. Available: <https://plg.uwaterloo.ca/~gvcormac/cormacksigir09-rrf.pdf>
- [14] S. Bruch, S. Gai, and A. Ingber, “An analysis of fusion functions for hybrid retrieval,” arXiv:2210.11934, 2022. [Online]. Available: <https://arxiv.org/abs/2210.11934>
- [15] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, and I. Androutsopoulos, “LEGAL-BERT: The muppets straight out of law school,” in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020. [Online]. Available: <https://aclanthology.org/2020.findings-emnlp.261/>
- [16] J. Zhan, J. Mao, Y. Liu, J. Guo, M. Zhang, and S. Ma, “Optimizing dense retrieval model training with hard negatives,” in *Proc. 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 1503–1512, 2021. [Online]. Available: <https://arxiv.org/abs/2104.08051>
- [17] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, “BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models,” in *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, 2021. [Online]. Available: <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/65b9eea6e1cc6bb9f0cd2a47751a186f-Abstract-round2.html>

- [18] K. Järvelin and J. Kekäläinen, “Cumulated gain-based evaluation of IR techniques,” *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002. [Online]. Available: <https://researchportal.tuni.fi/en/publications/cumulated-gain-based-evaluation-of-ir-techniques>
- [19] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” arXiv:1807.03748, 2018. [Online]. Available: <https://arxiv.org/abs/1807.03748>
- [20] American Statistical Association, “American Statistical Association releases statement on statistical significance and p-values,” 2016. [Online]. Available: <https://www.amstat.org/asa/files/pdfs/p-valuestatement.pdf>